Zhewen Shen

zhewen.shen@outlook.com
zhewenshen | zhewenshen

Education

UNSW Sydney
Bachelor, Advanced Computer Science (Honours) *Sep 2021–Dec 2025*

- Honours Class 1 and The University Medal
- Thesis: *Evaluating Programming Languages for Verified LionsOS Components*
  - Supervised by Scientia Professor Gernot Heiser

Work

Software Engineer Feb 2026–Present
Arista Networks

- Implemented anti-rollback protection for Arista EOS
- Brought up BIOS and bootloader support for products based on Intel Amston Lake platforms
- Engineered CPU–NIC protection by leveraging hardware-based packet filtering and extending kernel drivers to sustain line-rate 64-byte Ethernet traffic—14.88 Mpps at 10 GbE and 37.20 Mpps at 25 GbE
- Triaged software defects and customer escalations, diagnosing issues across EOS, the Linux kernel, device drivers, and platform firmware

Casual Academic May 2025–Present
School of Computer Science and Engineering, UNSW Sydney

- Teach classes for COMP6771 Advanced C++ Programming and COMP9242 Advanced Operating Systems
- Guide students through systems programming concepts, modern C++ features, and OS-level design principles
- Provide assignment feedback and support students during consultation hours

Research Intern (Trustworthy Systems) Jan 2025–Dec 2025
School of Computer Science and Engineering, UNSW Sydney

- Contributed to LionsOS, a component-based operating system built on the seL4 verified microkernel, designed for building custom, task-specific systems from composable components
- Implemented core migration and offlining mechanisms for multicore management in LionsOS
- Built a userspace webserver as a LionsOS component using its lock-free queue-based communication model
- Created a proof-of-concept transpiler to convert C code into Pancake, a research language with verified compilation for writing formally verified OS components
- Co-authored a paper at the Workshop on Programming Languages and Operating Systems (PLOS) 2026

Research Intern (NLP) Jan 2024–Jun 2024
School of Computer Science and Engineering, UNSW Sydney

- Developed BAMBINO-LM, a continual pre-training approach for small-scale language models inspired by bilingual language acquisition in children
- Combined language alternation strategies with PPO-based perplexity rewards to enable cross-lingual transfer between English and Italian
- Published at the Workshop on Cognitive Modeling and Computational Linguistics (CMCL) 2024

Skills

Languages: Advanced — **Keywords**: C, C++, Python, Bash
Tools & Infrastructure: Intermediate — **Keywords**: Git, Perforce, Docker, GDB, Make, CMake, Linux, MongoDB

Awards

The Jump Trading Hero of Operating Systems Prize Nov 2025
UNSW Sydney

The Optiver COMP6771 Advanced C++ Programming Prize Aug 2024
UNSW Sydney

Taste of Research Scholarship (2024 & 2025)
UNSW Sydney Faculty of Engineering